

Effects of soy as tofu vs meat on lipoprotein concentrations.

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OBJECTIVES: To investigate the effect of replacing lean meat with a soy product, tofu, on serum lipoprotein concentrations. STUDY AND DESIGN: Randomized cross-over dietary intervention study. SUBJECTS: Forty-two free-living healthy males aged 35-62 y completed the dietary intervention. Three additional subjects were non-compliant and excluded prior to analysis. INTERVENTIONS: A diet containing lean meat (150 g/d) was compared with one with 290 g/d tofu in an isocaloric and isoprotein substitution. Both diet periods were 1 month, and fat intake was carefully controlled. RESULTS: Seven-day diet records showed the two diets were similar in energy, macronutrients and fibre. Total cholesterol (mean difference 0.23 mmol/l, 95% CI 0.02, 0.43; P=0.03) and triglycerides (mean difference 0.15 mmol/l, 95% CI 0.02, 0.31; P=0.017) were significantly lower on the tofu diet than the lean meat diet. However, HDL-C was also significantly lower on the tofu diet (mean difference 0.08 mmol/l, 95% CI 0.02, 0.14; P=0.01) although the LDL-C:HDL-C ratio was similar. CONCLUSION: The effect on HDL-C and the small LDL-C reduction differ from some other studies, where fat was often less controlled, and the comparison was of soy as textured protein or soymilk against casein. This suggests a differential effect of the various proteins compared to the soy may influence the findings. In practice, the replacement of meat with tofu would usually be associated with a decrease in saturated fat and an increase in polyunsaturated fat and this should enhance any small benefits due to the soy protein. SPONSOR: Deakin University with some contribution from a Commonwealth Department of Veterans Affairs research grant. European Journal of Clinical Nutrition (2000) 54, 14-19